Laboratory Exercise 2: Sequential C Code

Learning Outcomes:

- Use input and output statement in C code.
- Explore line breaks, indentions and white spaces to format a C code properly
- Declare, initialize and assign values to variables in C
- Manipulate the values of variables using arithmetic operators
- Utilize built-in math functions for certain problems

Exercise Details

A sequential C code (without conditional and iterative structures) is the most fundamental means of solving programming problems. However, it is important to format the code properly in order to make it much readable. The three-step principle below will greatly improve the structure of your program

- One statement per line
- One line for the opening brace and one line for the closing brace
- All statements under a brace must be indented

For further details regarding the input/output statements, variables and datatypes, arithmetic operators and built-in math functions, please refer to your lecture manual.

Problems:

1. $r_{eff} = \left(1 + \frac{r}{m}\right)^m - 1$

   Where:

   $r = nominal\ interest\ rate\ (usually\ annual\ interest\ rate)$

   $m = number\ of\ conversion\ periods\ per\ year.$

   The formula above shows the rate of effective interest. Write a C code to compute the effective rate of interest given that *r* and *m* are input variables.

```
//output example:
Annual interest rate (%): 8
Number of interest conversions per year: 4
Effective interest rate is 8.243%
```

2. Using sequential programming in C (without conditional statements and/or iterative statements), write a program that would breakdown a positive integer input in pesos into its 1000's, 500's, 100's, 50's, 20's, 10's, 5's, 1's equivalent.
   Example:

```
1538 pesos
1:1000, 1:500, 0:100, 0:50, 1:20, 1:10, 1:5, 3:1
```

Write all you code inside function `main()`. Refer to the code formatting below:

```
#include <stdio.h>
#include <math.h>
int main()
{
      //Code for Problem 1

      //Code for Problem 2

}
```